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Design of Low Gain Avalanche Detectors (LGAD) with 400 keV ion implantation energy for multiplication layer fabrication

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ABSTRACT

Low Gain Avalanche Detectors (LGAD) are silicon sensors that can achieve a time resolution of better than 20 ps. The ATLAS and CMS experiments are designing LGAD detectors to address the pile-up challenge at the High Luminosity Large Hadron Collider (HL-LHC). The Institute of High Energy Physics (IHEP) has recently developed two versions of LGAD sensors. The LGAD sensors were designed using Technology Computer-Aided Design (TCAD) simulations and optimized to obtain high breakdown voltage and a suitable gain. The n-type Junction Termination Extension (N-JTE) and p-type gain layer are two critical structures for LGAD sensors that were investigated. IHEP has tuned the fabrication process of two foundries to obtain the most promising design. The first version of the IHEP LGAD sensor, with a gain higher than six and breakdown voltage higher than 400 V, was submitted to Tianjin Zhonghuan Semiconductor Company for fabrication. The second version of the LGAD sensor benefits from the higher implantation energy available at the Institute of Microelectronics (IME) to reach a gain higher than ten and breakdown voltage higher than 420 V.

1. Introduction

CERN will start the High Luminosity (HL) Phase-II of the Large Hadron Collider (LHC) in 2027 [1]. The HL-LHC will deliver an integrated luminosity of up to 4000 fb^{-1} over the subsequent decade. The instantaneous luminosity of the HL-LHC will reach up to 7.5×10^{34} $cm^{-2}s^{-1}$, a substantial increase from the 2.1×10^{34} $cm^{-2}s^{-1}$ obtained during LHC Run 2 [2,3]. The increase of instantaneous luminosity will increase the number of collisions in each bunch crossing. Improvement of the spatial and timing resolution is needed for particle tracks to be distinguished and primary vertices to be properly identified. One possible approach to satisfy these requirements is to use a tracking detector in conjunction with independent precise timing measurements. LGAD are silicon sensors that can achieve a time resolution of better than 20 ps. They have so far been developed by several silicon foundries [4–9]. The Institute of High Energy Physics (IHEP) High-Granularity Timing Detector group has recently developed its first two versions of LGAD sensors. In this paper, we report on the design, Technology Computer-Aided Design (TCAD) simulation and proposed fabrication technology for these LGAD sensors.

2. Structure

The LGAD structure drawn in Fig. 1 is based on the standard PIN diode architecture with an n^{++} layer as the cathode and a p^{++} layer as the anode. High voltage is applied on the anode and charge is collected from the n^{++} cathode at ground potential.

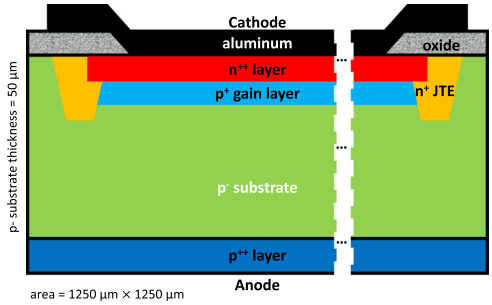


Fig. 1. Sketch of the LGAD structure with the active area shown. Height and width are not at the same scale. The sensor total area is 1250 μm × 1250 μm. The thickness of the p^- substrate is 50 μm.

The p^- substrate is the active volume for charge drifting. A thinner sensor contributes to better time resolution due to fewer Landau fluctuation. The proper gain factor can reduce jitter from electronic noise and time walk from amplitude variations allowing for high time

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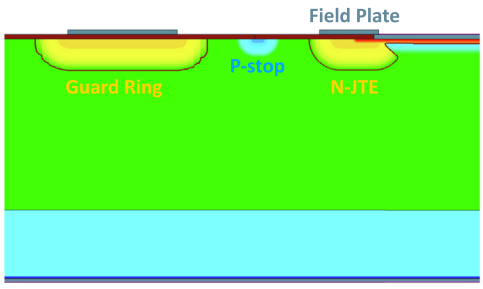


Fig. 2. Cross-section of the LGAD at the periphery showing doping concentrations of different regions.

resolution [2]. Under the n^{++} layer, a special gain layer composed of a p-type dopant delivers a low gain factor (10 ∼ 20). At the edge of the n^{++} and p^{+} regions a high electric field forms due to the beveled edge terminations. The high electric field will be improved by adding the Junction Termination Extension (JTE) [11] structure. The Junction Termination Extension (JTE) made by an n^{+} dopant can alter the surface electric field at the edges and increase the breakdown voltage. The JTE can also keep the active area electric field in better uniformity.

The p-stop [12] diffusion plotted in Fig. 2 can eliminate the electrical connection between n-type diffusions due to the surface inversion layer. The field plate structure is shown in Fig. 2 which extends the contact metal for the n++ electrode over the JTE. The field plate produces an extension of the depletion region along the surface which reduces the electric field at the JTE and increases the breakdown voltage. Guard rings are added at the device's periphery to shape the electric field distribution at the junction edges and shield the active area from the lateral leakage current [13].

3. TCAD simulation

Technology Computer-Aided Design (TCAD) [14] simulation of the electrical characteristics is the main method to predict the performance of real sensors. The electrical characterization simulation (with sdevice [14]) includes current–voltage (IV) simulation and gain simulation. The IV simulation provides the depletion voltage, leakage current, breakdown voltage, electric field distribution, and current density. The gain simulation based on a Minimal Ionizing Particle (MIP) signal provides the gain factor [10].

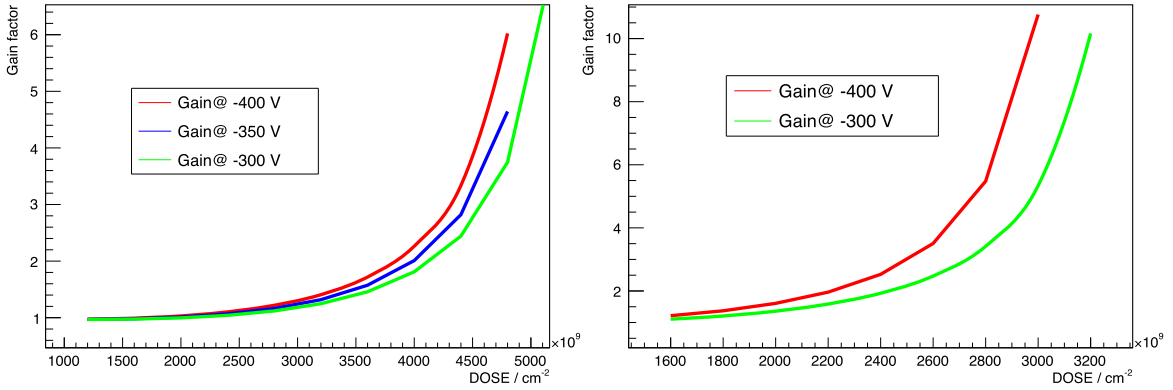


Fig. 3. Gain factor versus different ion implantation doses for different anode voltages. Left: A gain factor above six has been achieved with a dose range of 4.8 – 5.2 $\times 10^{12}$ cm^{-2} [10]. Right: A gain factor above ten has been achieved with a dose range of 3.0 – 3.2 $\times 10^{12}$ cm^{-2}. The left plot refers to the IHEP-Zhonghuan design and the right one to the IHEP-IME design.

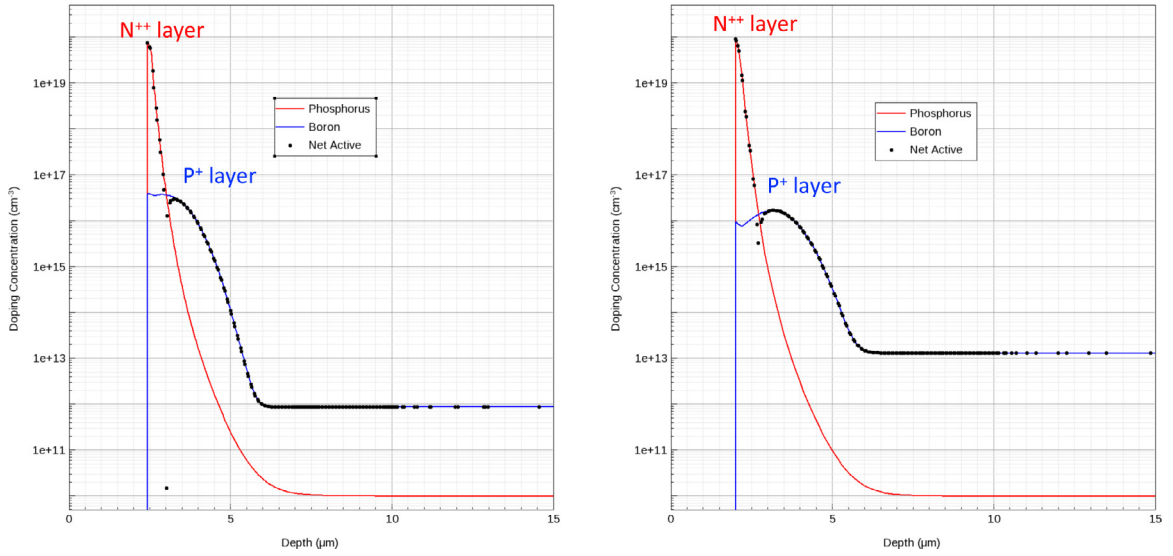


Fig. 4. N^{++} layer and p^{+} gain layer doping concentration distribution in different depths. The "Net Active" is the net active doping concentration after adding the n-type and p-type dopant concentration up. Left: In the IHEP-Zhonghuan version, the p^{+} gain layer was made by a 130 keV energy boron implantation process, the n^{++} layer was formed through the low energy phosphorus implantation process [10]. Right: In the IHEP-IME version, the p^{+} gain layer was made by a 400 keV energy boron implantation process, n^{++} layer was formed through the same low energy phosphorus implantation process as IHEP-Zhonghuan version's n^{++} layer. To produce the n^{++} layer in the shallow region above the gain layer, low energy (40 keV) phosphorus implantation is suitable.

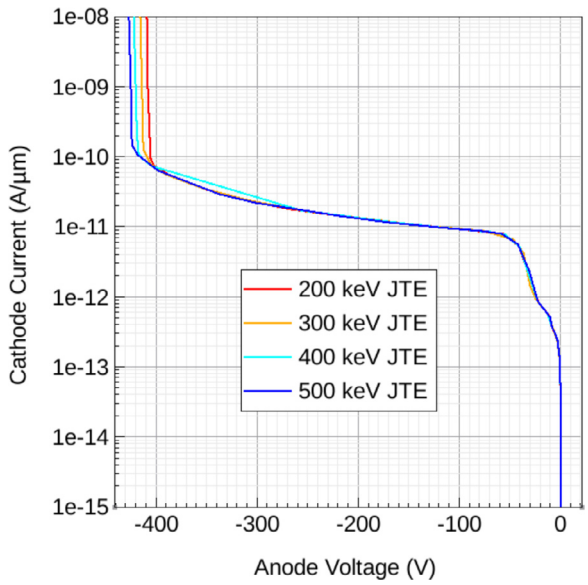


Fig. 5. IV simulation for different JTE implantation energies (200 keV, 300 keV, 400 keV, 500 keV).

4. IHEP-Zhonghuan LGAD design

IHEP has developed a first version LGAD sensor with Tianjin Zhonghuan Company. Optimization of LGAD design including structure optimization and doping profile optimization was mainly performed by TCAD simulation. After structure optimization including JTE, p-stop, field plate, and their distances, doping profile optimization was performed mainly on the gain layer fabrication process [10].

The final design for fabrication has a gain factor higher than six with a gain layer implantation dose higher than 4.8×10^{12} cm^{-2}. The gain at different bias voltages versus dose is shown in Fig. 3 left plot. The doping profile in different depths is shown in Fig. 4 left plot [10].

5. IHEP-IME LGAD design

IHEP has developed its second version LGAD sensor with the Institute of Microelectronics (IME). Compared with Zhonghuan, IME has different fabrication capabilities such as an 8-inch production line, broader lithography machine view field. The main improvement is the higher maximum energy of the ion implantation process. The maximum energy for phosphorus implantation is 600 keV (130 keV for Zhonghuan), the maximum energy for boron implantation is 400 keV (130 keV for Zhonghuan). The use of higher implantation energy benefits mostly the gain layer performance. The n^{++} layer was produced with low energy (40 keV) phosphorus implantation after the gain layer implantation. With all processes after gain layer implantation being the same, a higher gain could be achieved with lower implantation doses as Fig. 3 right plot shows. The dopants in the gain layer are more effectively activated through a better separation of the p^{+} gain layer and the n^{++} layer as Fig. 4 right plot shows. Compared to 130 keV implantation, the 400 keV implantation can drive boron into a deeper region. In the final design, a gain factor higher than ten could be achieved with a gain layer implantation dose higher than 3.0×10^{12} cm^{-2}.

Besides benefitting the gain layer in the active region, a higher ion implantation energy (600 keV for phosphorus) also allows the designer to tune the JTE process in a broader range. Fig. 5 shows that with a JTE implantation energy increase from 200 keV to 500 keV, the breakdown voltage increases from −400 V to −420 V. Due to the fact that the breakdown occurs at the JTE structure, changing the JTE shape can help increase the breakdown voltage. A JTE formed by higher implantation energy has a larger perimeter. The same potential change distributes over a wider JTE perimeter. The high electric field region around the JTE and the breakdown voltage are improved. Fig. 6 shows the change of the JTE doping profile with different implantation energies.

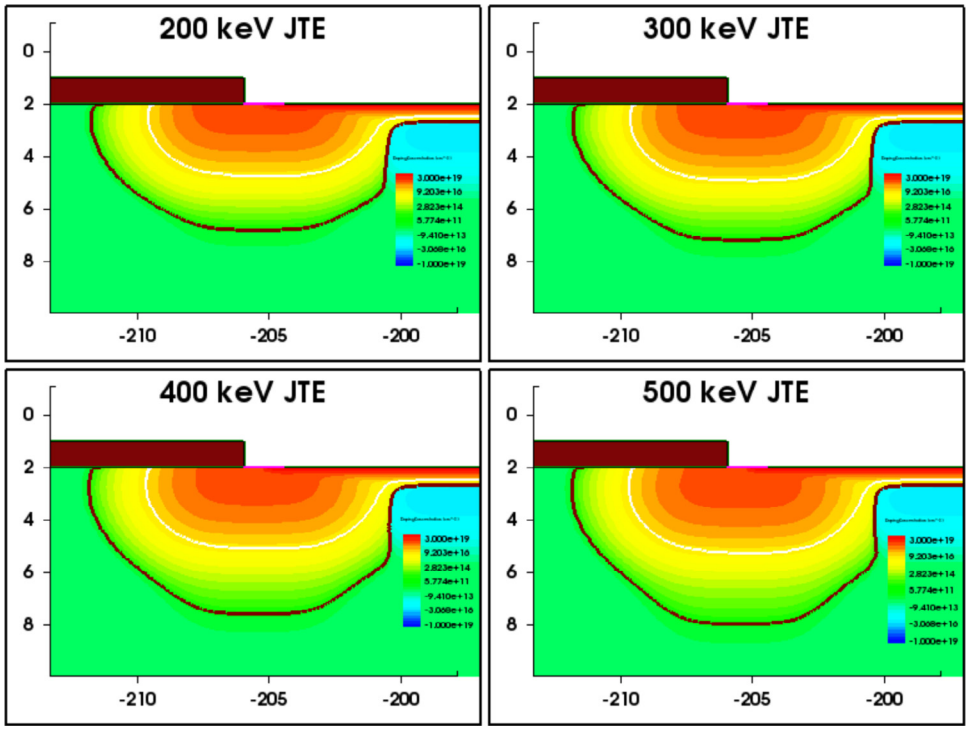


Fig. 6. Doping profile for different JTE implantation energies (200 keV, 300 keV, 400 keV, 500 keV). The white line is the boundary of the depletion region. The brown line is the junction line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6. Summary

Two versions of LGAD sensors have been designed, optimized, and submitted to two foundries: Tianjin Zhonghuan and IME. Although the fundamental designs are similar, IME offers the possibility of higher ion implantation energy. Taking advantage of this feature, the IHEP-IME LGAD sensors can achieve higher gain (10) with a lower boron implantation dose for the gain layer. The activated dopant percentage increases due to a better separation of gain layer and n^- layer. A higher breakdown voltage (−420 V) has been achieved by high energy phosphorus implantation for the JTE.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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